# Flow, Autonomy, and Organizational Heat: A Physical Systems Analogy for Psychology, Decision Architecture, and Organization Design

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March 15, 2026

**Abstract**

Organizational systems frequently exhibit patterns of friction, conflict, and systemic overload that resist explanation through conventional accounts of motivation, leadership, or communication. This paper develops a conceptual framework, grounded in hydraulic and electrical circuit analogies, to model how autonomy, legitimacy, capability, risk, and structural complexity interact to produce either smooth coordinated action or what we term *organizational heat*—the dissipative byproduct of action flow encountering structural resistance. The framework proposes five core formal relations as modeling hypotheses: autonomy as the product of legitimacy and capability; legitimacy and capability as bounded quantities subject to reduction by insignificance and risk, respectively; action flow as a ratio of autonomy to complexity resistance; and organizational heat as proportional to the square of flow times resistance. These algebraic forms are the authors' own formalizations by analogy to physical circuit theory and are not directly derived from the cited literatures, which are invoked to motivate the conceptual components rather than to validate the specific functional forms. The model is extended through Kirchhoff-like network constraints applied to decision architectures and through analysis of how AI-augmented organizations alter the flow–resistance–heat balance. We conclude with testable predictions that could validate or refute the framework's structural assumptions, and with an explicit assessment of model limitations.

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# 1 Introduction

## 1.1 The Problem of Organizational Instability

Organizations routinely experience coordination failures, decision overload, and escalating interpersonal conflict even when their members are individually competent and their objectives are clearly articulated. Standard accounts of such failures frequently invoke motivational deficits (Deci & Ryan, 1985), leadership dysfunction (Yukl, 2012), or communication breakdowns (Weick, 1995). While each of these accounts captures important phenomena, they do not adequately explain a class of instabilities that arise structurally—that is, as a consequence of how decision authority, information load, and interdependency are arranged, rather than as a consequence of individual deficiency or interpersonal failure.

Physical systems provide a potentially productive set of analogies for this structural class of problem. In fluid and electrical networks, flow is governed by driving potential and opposed by resistance; when flow encounters resistance, energy is dissipated as heat (Feynman et al., 1963). This paper proposes that analogous structural constraints may apply to the flow of decisions and actions within organizations. The specific algebraic forms through which we formalize this analogy—presented fully in Sections 2–6—are modeling choices rather than empirical findings; they are introduced because they generate tractable, falsifiable predictions, not because they are established by the literatures cited for motivation.

## 1.2 Literature Positioning

The framework draws on five bodies of established theory to motivate its conceptual components. We note explicitly where the framework is aligned with existing evidence and where it extends beyond what those sources directly support.

**Self-Determination Theory (SDT).** Deci and Ryan's self-determination theory (Deci & Ryan, 2000; Ryan & Deci, 2000) identifies autonomy, competence, and relatedness as fundamental psychological needs whose satisfaction reliably predicts intrinsic motivation and well-being across a wide range of empirical studies. This evidence base motivates our choice of autonomy as the primary driving variable in the model. The present framework departs from SDT in treating autonomy as a structural quantity with a specific algebraic form (the product of legitimacy and capability), rather than as a psychological need whose satisfaction falls on a continuous scale. That algebraic form is a modeling assumption, not something SDT or its associated empirical literature directly implies.

**Cognitive Load Theory (CLT).** Sweller's cognitive load theory (Sweller, 1988, 1994) establishes that working memory is a finite resource and that performance degrades when

total processing demand—across its intrinsic, extraneous, and germane components—exceeds available capacity. This finding supports the general principle that complexity imposes a cost on effective action, which motivates the resistance construct in the present model. The specific claim that organizational complexity resistance functions as the denominator in a ratio-form flow equation is our own formalization, not a conclusion of CLT.

**Systems Theory and Cybernetics.** General systems theory (Bertalanffy, 1968) and cybernetics (Ashby, 1956; Wiener, 1948) model organizations as open systems regulated by feedback, characterized by flows of energy, information, and material. This tradition motivates the use of flow and network concepts in describing organizations. The application of Kirchhoff-like conservation constraints to decision networks—developed in Section 7—is our extension of this tradition to the domain of authority and coordination flows; it is a conceptual analogy, not an established result in organizational systems theory.

**Control Theory and Normal Accidents.** Perrow's normal accident theory (Perrow, 1984) and Beer's viable systems model (Beer, 1972) both establish that organizational failure is systematically produced by structural mismatches between complexity and regulatory capacity. This provides empirical and theoretical grounding for the claim that structural properties, rather than individual behavior alone, determine system stability. The specific claim that this mismatch manifests as a quadratic heat function is our model's prediction, which requires empirical testing against the failure-mode patterns that Perrow and Beer document.

**Coordination Costs and Organization Design.** Transaction cost economics (Williamson, 1981) and coordination theory (Malone & Crowston, 1994) establish that organizational design choices have systematic effects on the costs of integrating distributed action. Puranam et al. (2012) demonstrate that coordination failure is predictably linked to specific structural properties of task interdependence. The present model builds on this tradition by treating coordination cost as a resistance parameter—a dynamic variable that interacts with autonomy to produce either stable throughput or dissipative heat. The specific functional form of that interaction is, again, our formalization.

## 1.3 Overview of the Framework

The model proposes the following formal relations as a coherent set of modeling hypotheses:

1. Autonomy as the product of legitimacy and capability
2. Legitimacy as a bounded quantity reduced by insignificance

3. Capability as a bounded quantity reduced by risk
4. Action flow as autonomy divided by complexity resistance
5. Organizational heat as the square of flow multiplied by resistance
6. Kirchhoff-like conservation constraints on decision networks
7. Decision architecture as the structural determinant of resistance

These relations form a self-consistent formal system. Their value as a theoretical contribution depends not on the precision of the analogy to physics but on whether they generate predictions that can be tested against organizational data. Section 12 develops these predictions explicitly.

Figure 1 presents the integrated model.

## 2 Autonomy as a Product of Legitimacy and Capability

$$\text{Autonomy} = \text{Legitimacy} \times \text{Capability}$$

Autonomy is defined in this framework as the effective capacity of an agent—individual, team, or organizational unit—to initiate and complete actions without requiring external authorization or compensatory support. This definition is broader than mere formal authority (permission to act) and requires both a recognized mandate and the practical ability to execute.

The multiplicative form is a deliberate modeling choice. It captures the intuition that effective autonomous action requires both factors to be substantially nonzero: an agent with full capability but no recognized mandate cannot act without authorization; an agent with full mandate but negligible capability cannot act reliably. A product structure formalizes this as a zero-condition: if either factor is zero, Autonomy is zero regardless of the magnitude of the other.

This choice is motivated by, but not directly implied by, the existing literature. Self-determination theory (Deci & Ryan, 2000) identifies autonomy and competence as distinct basic needs and documents their joint importance for intrinsic motivation; it does not, however, specify a multiplicative relationship between them at the organizational level. Research on role clarity (Rizzo et al., 1970) demonstrates that ambiguous authority reduces effective performance even when competence is high, which is consistent with the zero-condition property of the product form but does not uniquely identify it. The product structure should therefore be understood as a parsimonious formalization that is compatible with these findings, not as a conclusion that follows from them.

The multiplicative form also implies that marginal improvements in one factor yield diminishing returns when the other is low—a prediction that is consistent with the general

logic of complementarity in production (?) but that requires direct empirical validation in organizational settings.

# 3 Legitimacy and Insignificance

$$\text{Legitimacy} = \text{Legitimacy}_{\max} - \text{Insignificance}$$

Legitimacy denotes the degree to which an agent's authority to act is recognized and accepted by the relevant social system. The construct is related to but distinct from formal authority: legitimacy is conferred through relational and social processes, not solely through structural assignment. This corresponds to the social dimension of autonomy support documented in self-determination theory (Deci & Ryan, 2000) and to the "perceived organizational support" construct (Eisenberger et al., 1986), though neither source directly operationalizes legitimacy as formalized here.

$\text{Legitimacy}_{\max}$ represents the ceiling of authority available to the agent given their role and organizational position. Insignificance is a scalar construct introduced in this model to index the degree to which an agent's standing is undermined by structural or relational conditions. It is not a standard construct in the existing literature; the name and operationalization are proposed here as a modeling choice.

The following factors are proposed as antecedents of Insignificance, each supported by evidence that they reduce effective authority in practice:

- **Insufficient trust:** trust deficits are associated with increased reliance on formal monitoring and approval hierarchies (Mayer et al., 1995), which the model interprets as upward transfer of effective decision authority.
- **Mandate ambiguity:** role conflict and ambiguity empirically predict reduced performance and increased stress (Rizzo et al., 1970); the model formalizes this as a reduction in effective Legitimacy.
- **Low perceived status:** the mechanisms by which low status reduces an agent's ability to act without deference are well-documented in social influence research (Cialdini, 2001), though a direct mapping to Legitimacy as defined here remains to be established.
- **Absent accountability:** unclear ownership of outcomes is associated with diffuse responsibility and reduced initiative (Hackman & Oldham, 1980), which the model treats as a legitimacy deficit.

The linear subtraction form—where each unit of Insignificance directly offsets one unit of Legitimacy—is a simplifying assumption of the model. Real relationships between the

antecedents above and effective authority are likely nonlinear and context-dependent. This limitation is noted in Section 13.

# 4 Capability and Risk

$$\text{Capability} = \text{Capability}_{\text{max}} - \text{Risk}$$

Capability denotes the effective ability of an agent to execute actions reliably under prevailing conditions. The framing as a situationally bounded quantity reflects the well-established finding that performance depends not only on skill but on the demands imposed by the environment: the same agent may be highly capable under routine conditions and substantially impaired under conditions of high uncertainty, cognitive load, or consequence severity.

$\text{Capability}_{\text{max}}$ represents the agent's full repertoire of skills, knowledge, and resources under low-demand conditions. Risk in this model is not a property of outcomes per se but a demand on cognitive and attentional resources. This framing draws on Kahneman's model of attentional capacity (Kahneman, 1973), which establishes that higher-demand tasks reduce the resources available for concurrent activities, but extends it to the organizational level in a way that that source does not directly imply.

The following factors are proposed as determinants of the Risk demand variable:

- **Environmental uncertainty:** unpredictability increases monitoring demands and error-detection requirements (Perrow, 1984).
- **Task complexity:** higher intrinsic task complexity consumes working memory capacity (Sweller, 1994), leaving less available for productive execution.
- **Consequence severity:** high-stakes decisions direct attention toward loss-avoidance (Kahneman & Tversky, 1979), reducing resources available for efficient action generation.
- **Knowledge gaps:** operating under conditions of bounded rationality (Simon, 1972) increases the cognitive cost of each decision, which the model treats as a reduction in net Capability.

As with the Legitimacy equation, the linear subtraction form is a modeling simplification. The functional relationship between risk factors and effective capability is almost certainly nonlinear in practice, and the specific choice of linear subtraction is made for analytical tractability. The parallel algebraic structure of the Legitimacy and Capability equations is also a deliberate formal symmetry: both components of Autonomy are modeled as bounded resources subject to environmental depletion. Whether this symmetry

reflects a deeper empirical regularity or is merely a convenient formal property of the model is an open question.

# 5 Action Flow

$$F = \frac{\text{Autonomy}}{\text{Resistance}_{\text{Complexity}}}$$

Action flow $F$ represents the rate at which decisions and coordinated actions are generated and completed per unit time within the organizational system. The ratio form is directly inspired by Ohm's law in circuit theory (Feynman et al., 1963), where current equals voltage divided by resistance. This is an analogy, not a derivation: there is no prior theoretical or empirical result that directly implies this functional form for organizational systems. The ratio is proposed because it generates clear, testable predictions about the effects of autonomy expansion and resistance reduction as independent and interacting interventions.

$\text{Resistance}_{\text{Complexity}}$ subsumes all structural and informational factors that impede the throughput of decisions and actions. The following categories of resistance are proposed, each supported by evidence that they increase coordination cost or reduce decision throughput:

- **Coordination requirements:** the need to achieve alignment among multiple actors before action can proceed has been established as a systematic source of organizational cost (Malone & Crowston, 1994).
- **Information overload:** processing demand in excess of working memory capacity degrades decision quality and speed (Sweller, 1988).
- **Constraint conflict:** simultaneous obligations that cannot all be satisfied by a single decision force negotiation or escalation, both of which consume resources that would otherwise support throughput.
- **Structural opacity:** ambiguous authority maps and missing escalation paths are documented sources of coordination failure (Mintzberg, 1979).

The ratio form implies that flow can be increased either by increasing Autonomy or by decreasing Resistance. These are architecturally distinct interventions with different cost profiles. However, the critical implication of the ratio form—in combination with the heat equation—is that high Autonomy under high Resistance does not produce high stable flow; it produces high organizational heat (Section 6). Whether this prediction holds quantitatively, as opposed to qualitatively, is an empirical question the model cannot answer from its assumptions alone.

# 6 Organizational Heat

$$H = F^2 \times \text{Resistance}$$

Organizational heat $H$ is the central novel construct of this framework. It is defined as the rate of unproductive dissipation within the organizational system—manifesting as stress, interpersonal conflict, rework, escalation, and decision fatigue. The term and its formal expression are direct analogies to Joule heating in electrical systems (Feynman et al., 1963), where power dissipation equals the square of current times resistance. This is a conceptual innovation of the present model and is not a standard construct in any of the literatures cited for motivation.

The quadratic dependence on flow is the most consequential structural feature of the model. Under this formulation, doubling flow while holding resistance constant quadruples heat. This non-linearity generates a specific prediction: systems that operate stably under moderate action rates may exhibit abrupt, disproportionate degradation under higher loads. This qualitative pattern is documented across several literatures. The Yerkes–Dodson relationship between arousal and performance (Yerkes & Dodson, 1908) establishes an inverted-U pattern in which high arousal impairs performance—consistent with the prediction that high flow generates costs—but does not imply a quadratic relationship specifically. Perrow's analysis of tightly coupled systems (Perrow, 1984) documents how small perturbations can cascade to catastrophic failure, which is consistent with superlinear heat accumulation but is not formalized in those terms. The specific quadratic form of the heat equation is a modeling assumption that warrants direct empirical investigation.

Substituting the flow equation yields the alternative expression:

$$H = \frac{\text{Autonomy}^2}{\text{Resistance}_{\text{Complexity}}}$$

This form reveals an important design asymmetry: under the model's assumptions, increasing Autonomy without reducing Resistance raises heat superlinearly, while reducing Resistance (holding Autonomy constant) reduces heat. The practical implication is that resistance reduction is a more leverage-efficient intervention than autonomy expansion under high-complexity conditions. This is a prediction of the model, not an established finding.

When heat is not dissipated through structural adjustment, the model predicts accumulation as latent instability. The psychological correlates proposed—burnout (Maslach & Jackson, 1981), trust erosion (Mayer et al., 1995), and risk aversion—are all empirically documented outcomes of sustained organizational overload. The model proposes that they share a common structural cause expressible through the heat equation; whether that causal unification is valid is an empirical question.

# 7 Kirchhoff Constraints in Decision Networks

The scalar relations of Sections 5–6 describe individual agents or units in isolation. Real organizations are networked: decisions depend on prior decisions, authorities overlap, and information flows along paths. To model these network effects, we introduce Kirchhoff-like conservation constraints by direct analogy to Kirchhoff's current and voltage laws in electrical network theory (Feynman et al., 1963).

These constraints are the authors' extension of the flow analogy to networked systems. They are not derived from organizational theory but are proposed as useful formal tools for identifying structural sources of coordination failure. Their value depends on whether the predictions they generate— developed in Section 12—are empirically supported.

## 7.1 Node Conservation: Flow Balance

We propose that at any decision node $n$ in the organizational network, a conservation-like constraint should hold between incoming and outgoing decision flow:

$$\sum_i F_{\text{in},i} = \sum_j F_{\text{out},j}$$

A decision node represents any locus where multiple information streams, authorities, or constraints must be reconciled before an output decision can be produced. Common examples include a project manager integrating inputs from engineering, finance, and compliance before committing to a delivery plan, or a cross-functional committee reconciling competing resource claims.

The proposed conservation principle states that if incoming flows consistently exceed the node's processing capacity, the excess accumulates as a backlog of unresolved decisions. This backlog is itself a source of increased Resistance for subsequent flow, creating a positive feedback: backlog increases resistance, which reduces effective throughput, which may increase heat as upstream agents attempt to force decisions through the congested node. This feedback structure is qualitatively consistent with documented patterns of decision escalation and overload in management research (Sterman, 1989), but the formal correspondence has not been empirically established.

Decision nodes must reconcile all incoming constraints simultaneously, which may include strategic goals, resource limits, risk and compliance policies, stakeholder demands, and temporal dependencies. When the incoming constraint set is overdetermined—when no single decision can simultaneously satisfy all constraints—the node cannot produce a valid outgoing flow. The model characterizes this as a node-level conservation violation. In practice, such violations manifest as decision deadlock or escalation; the model predicts that nodes with chronically overdetermined constraint sets will be persistent heat sources. This prediction is testable and is stated formally in Section 12.

## 7.2 Loop Constraints: Potential Balance

For closed loops within the decision network, we propose the following constraint by analogy to Kirchhoff's voltage law:

$$\sum_{\text{loop}} \text{Autonomy}_k = \sum_{\text{loop}} F_k \times \text{Resistance}_k$$

This constraint states that the total driving potential invested in a decision cycle must be fully accounted for by productive work done against resistance within that cycle. When the balance is violated—when more potential is invested than the cycle can convert to output—the residual accumulates as heat or forces revision of the assumptions underlying the cycle.

Organizational loops subject to potential imbalance include:

- **Strategy–execution–evaluation–strategy cycles:** where strategic ambition is not matched by the capability and resistance profile of the execution layer, producing persistent performance gaps that the feedback mechanism cannot correct (Sterman, 1989).
- **Policy–implementation–reporting–policy cycles:** where compliance reporting is structurally decoupled from implementation realities, producing cycles that systematically miscalibrate the policy layer.
- **Objective–action–feedback–objective cycles:** where feedback latency or distortion prevents accurate potential signals from reaching the goal-setting process.

## 7.3 Coordination Failures as Conservation Violations

The framework proposes that coordination failures can be formally characterized as violations of node or loop conservation. A unit that generates more outgoing decisions than its incoming authority can support is operating with a legitimacy deficit. A unit that absorbs incoming decision flow without producing proportionate outputs is generating a resistance backlog. Both conditions degrade systemic flow efficiency and increase total organizational heat.

This characterization has diagnostic value: it suggests that persistent coordination failures can be localized to specific nodes or loops where conservation is violated, and that structural interventions targeting those points should be more effective than diffuse cultural or motivational interventions. Whether this diagnostic strategy outperforms alternatives in practice is an empirical question that the present framework cannot resolve on theoretical grounds alone.

# 8 Decision Architecture

Decision architecture is defined in this framework as the structural configuration that determines how decisions are generated, routed, authorized, and reconciled within an organizational system. It is the primary design lever for $\text{Resistance}_{\text{Complexity}}$ and, through the Kirchhoff constraints, for whether node and loop conservation can be satisfied.

## 8.1 Relationship to Governance, Management, and Control Systems

Decision architecture, as defined here, differs from governance, management, and control systems in its level of abstraction. Governance specifies who has ultimate authority; management specifies how that authority is exercised; control systems specify how deviations from desired states are detected and corrected. Decision architecture specifies the structural topology through which all three operate—the network of decision rights, information channels, and constraint relationships that determines resistance.

This construct is conceptually related to Beer's viable systems model (Beer, 1972), in which system-level cohesion depends on the regulatory structure of the metasystem. It is also related to Williamson's treatment of organizational form as a determinant of transaction costs (Williamson, 1981). The present model extends these frameworks by treating the architecture as a determinant of a dynamic resistance variable rather than as a source of static coordination costs. Whether this dynamic extension adds predictive value over static coordination-cost accounts is an open empirical question.

## 8.2 Structural Sources of Resistance

The following structural properties are proposed to increase Complexity Resistance, in each case with reference to evidence that they impose coordination costs:

- **Authority ambiguity:** role conflict and boundary ambiguity are empirically associated with performance decrements and stress (Rizzo et al., 1970); the model interprets this as a resistance increase.
- **Responsibility overlap:** diffusion of accountability is associated with reduced individual effort (Latané et al., 1979); the model treats this as a coordination friction.
- **Information–authority separation:** when decision rights are held by agents who lack the specific knowledge required to exercise them, information must be transferred before action can proceed (Jensen & Meckling, 1992); this transfer cost enters as resistance.
- **Incentive misalignment:** divergent incentive structures among interacting agents generate negotiation costs at decision nodes (Holmström & Milgrom, 1991).

- **Feedback latency:** delayed feedback is a documented source of oscillation and instability in decision systems (Sterman, 1989); the model treats it as a resistance component that increases the probability of constraint violations.

### 8.3 Principles of Low-Resistance Architecture

From the model's assumptions, the following design principles reduce Complexity Resistance and are therefore predicted to improve flow stability and reduce heat:

1. Assigning clear, non-overlapping decision authority for each recognized decision class.
2. Collocating authority with the specific knowledge required to exercise it (Jensen & Meckling, 1992).
3. Designing feedback mechanisms with sufficiently low latency to support the required throughput rate.
4. Resolving constraint conflicts at the architectural design level, so that individual decision nodes receive consistent, satisfiable constraint sets.
5. Limiting the number of simultaneous incoming constraints at any node to within its estimated processing capacity.

These are predictions of the model. The general principles are consistent with established design guidance (Mintzberg, 1979; Williamson, 1981), but the specific prediction that violating them generates superlinear heat—rather than merely linear cost—is a novel claim that requires empirical support.

## 9 AI-Native Organizations

### 9.1 How AI Augments Potential Flow

AI-augmented organizations—systems in which artificial intelligence performs a substantial fraction of information processing, decision-support, or execution functions—alter the model's parameters in ways that are qualitatively distinctive. There is empirical evidence that AI tools extend the range of tasks that can be executed reliably (Agrawal et al., 2018) and reduce the time required per decision, which in the model's terms increases Autonomy at the agent or unit level by reducing Risk demand on human Capability. Automation similarly reduces variance in routine execution, decreasing the uncertainty component of Risk.

Under the model's flow equation $F = \text{Autonomy}/\text{Resistance}$, these effects produce a substantial increase in potential action flow. Whether that potential is realized as productive output or dissipated as heat depends on whether Resistance is adjusted proportionately. The model's prediction—and this is explicitly a model-derived prediction rather than an observed regularity—is that when AI increases Autonomy without a proportionate reduction in Resistance, the heat equation $H = F^2 \times \text{Resistance}$ predicts superlinear heat accumulation.

## 9.2 Why Heat May Increase Faster Than Flow

The model predicts that AI-augmented systems face an asymmetric risk: Autonomy-increasing mechanisms and Resistance-reducing mechanisms operate independently and are not automatically co-produced. Several properties of AI systems are proposed to generate resistance of their own:

- **Objective misalignment:** AI systems optimizing proxy metrics can diverge from organizational intent, creating constraint conflicts at integration nodes (Russell, 2019). The model treats each such conflict as a Kirchhoff node violation.
- **Metric proliferation:** multiple AI subsystems generating competing performance signals increase constraint overdetermination at human oversight nodes.
- **Opacity and legitimacy loss:** non-interpretable AI decisions may reduce the social legitimacy of AI-generated authority within the human organizational system, which the model formalizes as an increase in Insignificance for AI-origin flows.
- **Pace asymmetry:** AI subsystems may generate decisions at rates that exceed the processing capacity of human governance nodes, producing backlogs that the model identifies as conservation violations.
- **Cascading dependencies:** tightly coupled AI pipelines increase the topological complexity of the decision network, multiplying the number of Kirchhoff constraints that must be simultaneously satisfied.

These effects are proposed as mechanisms by which AI integration could generate heat faster than flow. Whether they do so in practice, and whether the magnitude follows the quadratic prediction, are empirical questions that organizational research on AI-augmented work could in principle address.

## 9.3 Architectural Requirements for AI-Augmented Stability

The model generates the following specific structural predictions for AI-augmented organizations:

- Each AI agent requires a clearly specified authority domain (legitimacy), with non-overlapping claims relative to other agents and to human decision-makers.
- The objective functions of interacting AI agents must be mutually consistent to avoid loop-level potential imbalance.
- Human oversight nodes must have sufficient processing capacity to handle the flow generated by AI subsystems, or AI decision rates must be deliberately throttled to match human governance bandwidth.
- Feedback latency must be low enough to detect objective drift and constraint conflicts before they accumulate as unresolved backlogs.

These are architectural prescriptions derived from the model's assumptions. They are consistent with the governance recommendations of Russell (2019) and with emerging frameworks for AI safety and controllability, but their specific derivation from the heat equation is novel to this framework.

# 10 The Coordination Economy

## 10.1 Coordination Cost as the Binding Constraint

The term *coordination economy* is used here to describe a class of productive environment in which the technical means of production are not the binding constraint on value creation, and coordination among distributed agents constitutes the primary bottleneck. This environment is characteristic of knowledge-intensive industries, platform ecosystems, and multi-agent AI systems. The shift from production cost to coordination cost as the binding constraint in knowledge work is documented in the organizational and platform economics literature (Malone & Crowston, 1994; Parker et al., 2016; Puranam et al., 2012).

## 10.2 Why Flow, Resistance, and Heat Matter at Scale

In low-complexity, low-flow environments, the heat generated by suboptimal decision architecture is modest. As scale increases, however, the quadratic heat equation becomes the dominant structural constraint on organizational viability—under the model's assumptions. To see this, consider two organizations with identical Autonomy levels but resistance profiles $R$ and $2R$ respectively. If both operate at flow $F$, their heat differential is:

$$H_B - H_A = F^2 R$$

This differential grows with the square of flow, meaning that architectural advantages compound significantly as scale increases. Organizations with lower resistance can sus-

tain higher throughput before encountering the heat threshold that produces systemic instability.

The model therefore predicts that in coordination-economy environments, organizations that minimize resistance through architectural design will realize compounding coordination advantages—but this is a prediction of the model, derived from the heat equation, not an empirically established regularity. The coordination economics literature establishes that coordination costs matter; the specific prediction that they scale quadratically with throughput is a novel theoretical contribution that requires empirical validation.

Under these conditions, the model predicts that:

- Autonomy expansion that outpaces Legitimacy and Capability development will generate heat rather than value.
- Resistance reduction through architectural redesign will produce compounding returns at scale.
- Kirchhoff constraint maintenance becomes an increasingly binding organizational discipline as network size grows.
- Heat monitoring should precede, not follow, the behavioral and performance outcomes it produces.

## 11 Psychological Interpretation

The model generates predictions about individual psychological states as functions of the structural variables. These predictions connect the model's constructs to well-established empirical findings, but the specific causal path proposed—from structural variables to heat to psychological state—is the model's hypothesis, not a claim directly supported by the cited sources.

- **Low Legitimacy (high Insignificance):** when an agent's organizational standing is systematically undermined, effective Autonomy approaches zero. The model predicts behavioral passivity; the corresponding empirical construct is learned helplessness (Seligman, 1975)—a state in which prior non-contingent outcomes suppress future initiative. The structural parallel is suggestive, though the original helplessness research was conducted in individual learning contexts, not organizational ones.
- **High Risk (low effective Capability):** when perceived environmental demands exceed effective competence, the model predicts aversive arousal; the empirical correlate is anxiety (Lazarus & Folkman, 1984). Again, the structural prediction is consistent with this finding, but the path from the model's formal variables to the psychological outcome requires empirical mediation analysis to establish.

- **High Flow with High Resistance:** sustained action output against high structural friction is predicted by the heat equation to generate rapid energy dissipation. The empirical correlate is burnout (Maslach & Jackson, 1981). The demand–resource framework in burnout research (?) offers a closely parallel structural account in which demands exceeding resources produce exhaustion, lending theoretical support to this mapping.

- **Balanced Flow:** when Autonomy is matched to Resistance, flow is stable and productive. The predicted psychological correlate is Csikszentmihalyi's flow state (Csikszentmihalyi, 1990), in which challenge is matched to skill and action proceeds effortlessly. The terminological overlap between "action flow" in this model and "flow" in Csikszentmihalyi's sense is intentional but does not imply that the constructs are identical.

These correspondences suggest that the framework may bridge structural organizational variables and psychological outcomes within a common formal language. That bridge requires empirical support at each link in the proposed causal chain.

# 12 Testable Predictions

The framework generates the following empirically testable predictions. Each is stated in a form that specifies the functional relationship and the conditions under which it would be falsified.

1. **Superlinear heat scaling:** Organizational stress indicators (absenteeism, conflict frequency, decision latency, escalation rates, self-reported exhaustion) will increase superlinearly with action throughput when structural resistance is held approximately constant. The specific functional form predicted is quadratic. A linear relationship would falsify the $H = F^2R$ formulation and support a simpler model.

2. **Resistance as a moderator of autonomy interventions:** Organizational interventions that expand agent autonomy (role empowerment, delegation, reduced approval requirements) will produce positive performance outcomes in low-resistance contexts and null or negative outcomes in high-resistance contexts, where the model predicts that increased autonomy generates heat rather than flow.

3. **Overdetermination as a conflict predictor:** Decision nodes at which the number of simultaneously active constraints exceeds some threshold will exhibit higher rates of escalation, conflict, and decision avoidance than nodes operating below that threshold, controlling for other organizational characteristics.

4. **AI-induced heat in under-architected systems:** Organizations introducing AI systems that substantially increase decision throughput without proportionate architectural redesign will exhibit increased organizational heat indicators, measurable as increased coordination overhead, role conflict, and decision reversal rates, relative to organizations that co-invest in architectural adaptation.

5. **Legitimacy–Capability asymmetry in intervention effectiveness:** Competence development interventions (training, tools) applied in low-legitimacy contexts will not produce proportionate improvements in effective Autonomy, because the multiplicative form implies that Capability gains are multiplied by a near-zero Legitimacy factor. Conversely, authority expansion in low-Capability contexts is predicted to increase Risk demand (by raising the stakes of action) without a proportionate gain in effective Capability, producing net heat.

6. **Loop drift from potential imbalance:** Decision cycles in which the driving potential—measured as invested effort, authority commitment, and resource allocation—is consistently mismatched to outputs will exhibit systematic drift in cycle time, revision frequency, and strategy–execution divergence, at rates exceeding those observed in cycles where potential and output are balanced.

# 13 Limitations of the Model

The following limitations are inherent to the framework as currently formulated and should be addressed in any research program designed to evaluate its claims.

1. **Unvalidated measurement:** The model's constructs—Legitimacy, Capability, Insignificance, Risk, and Resistance—are defined conceptually but do not yet have validated psychometric or organizational measurement instruments. The algebraic precision of the expressions is not currently matched by corresponding precision in operationalization. Construct validation studies are a necessary precondition for empirical testing of any of the predictions in Section 12.

2. **Assumed functional forms:** The specific functional forms—multiplicative for Autonomy, linear subtraction for Legitimacy and Capability, ratio for Flow, and quadratic for Heat—are modeling choices made for analytical tractability and conceptual coherence. Empirical relationships among the underlying variables may be nonlinear, discontinuous, threshold-dependent, or interaction-dependent in ways that these forms do not capture. The model makes no claim that these are the uniquely correct functional forms.

3. **Static structure:** The framework describes system state at a point in time. Organizational systems are dynamic: resistance changes as architecture adapts, legitimacy changes as relationships develop, and capability changes through learning and attrition. The model does not currently specify differential equations or feedback dynamics governing these changes.

4. **Interpersonal and cultural factors:** The framework models organizations as flow networks and does not represent interpersonal relationships, organizational culture, or historical path dependence— all of which are empirically significant (Schein, 1985). The model's resistance variable is intended to absorb some cultural effects, but this substitution is not theoretically grounded.

5. **Kirchhoff analogy limitations:** Kirchhoff's laws are formally valid for linear, time-invariant networks. Organizational decision networks are nonlinear, time-varying, and subject to exogenous shocks. The conservation constraints introduced in Section 7 should be understood as qualitative structural hypotheses rather than precise quantitative laws.

6. **Endogeneity:** The model specifies a causal structure in which architecture determines resistance, which determines flow and heat. In practice, heat outcomes (conflict, burnout, turnover) alter architectural properties, creating feedback loops not currently represented in the model. Causal identification in any empirical test will need to address this endogeneity through longitudinal or experimental designs.

7. **The "heat" metaphor and reification risk:** The term organizational heat carries strong inferential connotations from physics that may not transfer reliably to organizational contexts. Readers and researchers should treat the term as a label for a proposed latent variable—unproductive dissipation under structural load—rather than as a claim that organizations literally obey thermodynamic laws.

# 14 Conclusion

This paper has proposed a structural model of organizational dynamics grounded in physical flow analogies and positioned within established theoretical traditions in psychology, systems theory, and organization design. The core framework formalizes Autonomy as the product of Legitimacy and Capability, Action Flow as the ratio of Autonomy to Complexity Resistance, and Organizational Heat as a quadratic function of Flow and Resistance. These are modeling hypotheses—deliberately stated in formal terms to enable testable predictions—not summaries of existing findings.

The framework is motivated by, but distinct from, its theoretical antecedents. Self-determination theory, cognitive load theory, cybernetics, normal accident theory, and co-

ordination economics each supply evidence for one or more of the model's conceptual components; none of them directly implies the specific algebraic forms proposed here.

The model's primary theoretical contributions are threefold. First, it proposes a unified formal language for phenomena—learned helplessness, burnout, coordination failure, escalation overload—that are typically treated in separate literatures. Second, it applies Kirchhoff-like conservation constraints to decision networks, generating structural predictions about where coordination failures will occur and why they are stable. Third, it applies the flow–resistance–heat framework to AI-augmented organizations, generating predictions about the conditions under which AI-driven capability expansion produces organizational instability rather than productivity.

The testable predictions developed in Section 12 provide a concrete research agenda. Testing them would require both construct validation—developing reliable measures for the model's latent variables— and structural equation modeling or experimental designs capable of estimating the functional forms proposed. Until such tests are conducted, the framework should be understood as a well-structured theoretical proposal whose value lies in the precision and falsifiability of its claims, not in their prior empirical confirmation.

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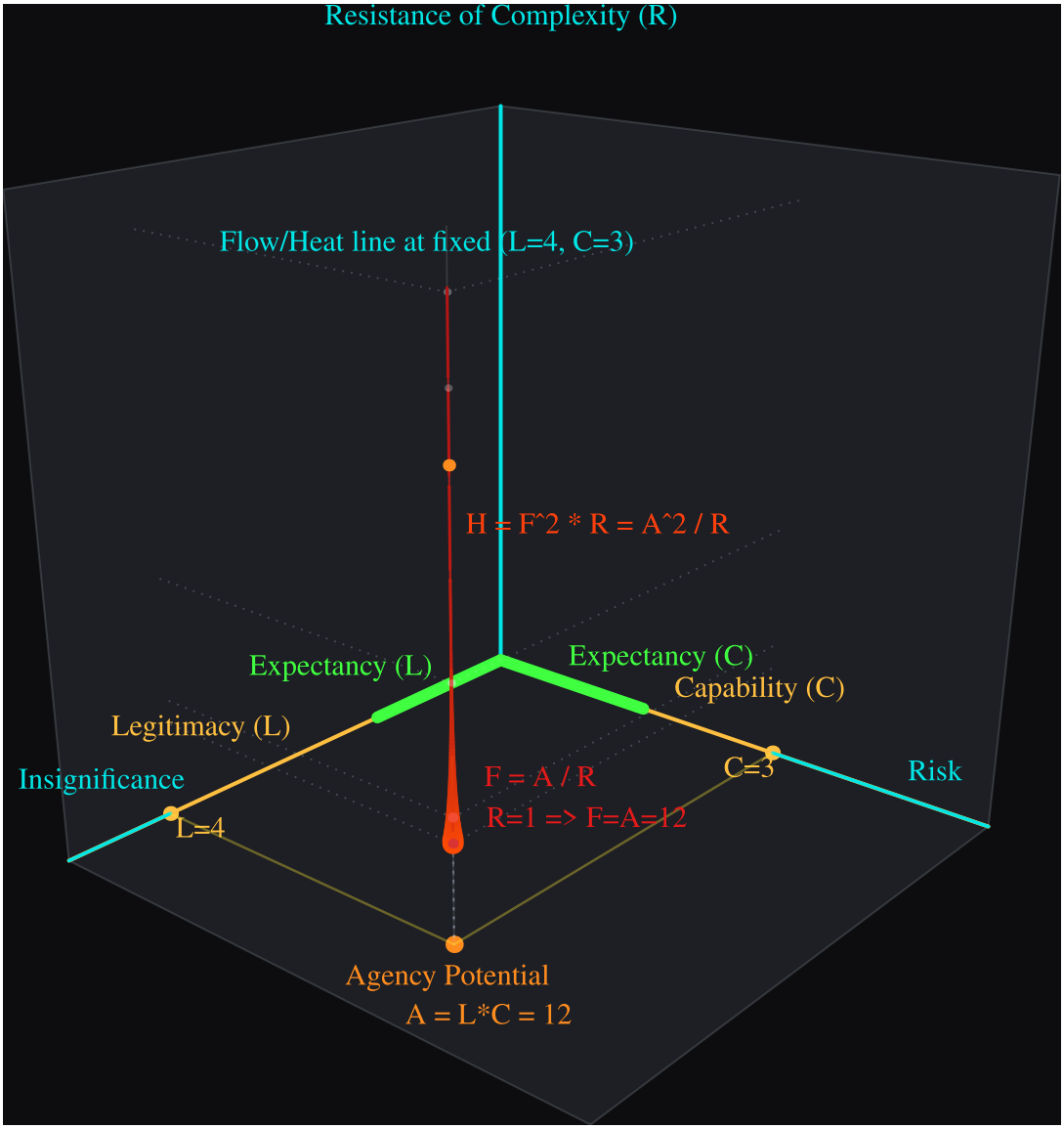


Figure 1: Integrated diagram of the Autonomy–Flow–Heat framework. Arrows denote the causal dependencies proposed by the model. The structure mirrors an electrical circuit with a driving potential and resistive dissipation, but the mapping is analogical; the diagram is not intended to imply that organizational quantities are measurable in physical units.